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THE MSISSI NORITE REVISITED : K/AR DATING, PETROGRAPHY AND PALEOMAGNETISM.

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Abstract. The "Msissi norite" in Morocco (30.93°N, 4.71°W) from where one of the paleomagnetic reference pole for Africa in Devonian times has been derived, has been revisited and dated. Petrological examination shows that the rock is an alkaline gabbro (teschenite). K-Ar analyses on fresh biotite yield a 136-139 Ma age, which infirms the supposed Devonian age. The paleomagnetic study displays several distinct magnetic components.

Introduction

The Msissi norite, paleomagnetically investigated by Hailwood (1974), has been heretofore regarded as Late Devonian in age on the basis of rather tenuous field arguments, and constitutes the only paleomagnetic reference pole (0.5°S ; 25°E) of that age for Africa. These data are crucial to the possible existence of a wide ocean separating Laurasia from Gondwana at that time. Several recent studies, Bachtadse et al. (1984) or Hurley and Van der Voo (1985) support Hailwood's results whereas those obtained by Kent et al. (1984) are in contradiction. Our purpose was first to check the absolute age of this pole, and second, to complete the paleomagnetic study.

Geology

The Msissi norite (location 30.93°N, 4.71°W) (Figure 1) outcrops in the Maïder (Eastern Anti-Atlas), 60 km south of the High-Atlas's southern limit, 35 km to the south-south east of the Msissi village. The pluton intrudes limy and pelitic Famennian and Tournaisian formations, developing a narrow contact metamorphism aureole (a few meters wide). The outcrop, 5 km long over 2 km wide, seems to be part of a massive dyke which extends on about 10 km. It is situated on the eastern limb of a very broad syncline. In the field, there is no indication of regional metamorphism nor deformation.

Petrology

The examination of thin sections reveals that the main mineralogical components of this dark coloured rock are : - plagioclase - clinopyroxene - abundant biotite - scarce amphibole and titanomagnetite, with a doleritic structure. Being partly damouritised, plagioclase has an anorthite content between 3 and 58%. The composition of titaniferous clinopyroxene falls near the salite-augite boundary. Brown kaersutitic amphibole and biotite surround the clinopyroxene and have partly replaced this mineral during the last magmatic stage. Biotite is generally automorphous. Some of them underwent a slight vermiculite-chlorite transformation. But the chemical compositions of fresh biotites are consistent (table 1), and rather normal for a biotite. The material is highly suitable for K-Ar dating. Analcite, vesicles of prehnite and zeolithe, present as accessory minerals, indicate with the biotite-vermiculite transformation, that a low hydrothermal stage immediately followed the last magmatic stage. The presence of kaersutite, titaniferous salite-augite, biotite, titanomagnetite, zeolite and analcite determined by microprobe shows that this gabbro is typically alkaline and therefore be called teschenite.

Age determination

All the K-Ar determinations were performed on separated minerals. The 100-160 μm sieve fraction of the ground rock samples was chosen for magnetic separation of biotites. Whenever necessary, further refinement was done with heavy liquids. The analytical procedures were described in detail in Westphal et al. (1979). Potassium was measured in a flame photometry with lithium internal standard. Argon was extracted in a bakeable glass vacuum apparatus and determined by isotope dilution technique using ^{38}Ar as a tracer with a MS10 mass spectrometer. All samples were treated in static mode. The results and the estimates of analytical precision at one standard deviation are listed in table 2. They were calculated following the procedure given in Cox and Dalrymple (1967), using the set of constants recommended by Steiger and Jaeger (1977). Ages yielded by fresh biotites are consistent between 136-139 Ma. Given the relative moderate size of

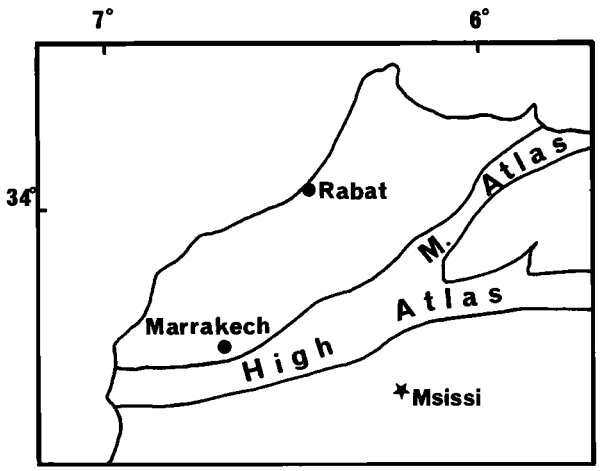


Fig.1. Localization of the Msissi village (30.93°N, 4.71°W).

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TABLE.1 Biotite analysis.

N°	6 Nm10	24 Nm10	33 Nm10	1 Nm5
SiO_2	34.48	36.7	36.32	37.12
TiO_2	5.75	7.37	7.46	6.81
Al_2O_3	14.79	13.73	13.21	13.88
FeO	23.96	19.15	17.95	18.87
MnO	0.17	0.03	0.19	0.17
MgO	8.05	11.02	11.34	11.73
CaO	0	0.08	0	0
Na_2O	0.42	0.3	0.78	0.83
K_2O	8.86	8.77	8.7	8.38
NiO	0.02	0	0	0.04
ZnO	0	0	0	0
Cr_2O_3	0.05	0.03	0	0
	96.55	97.19	95.95	97.84
Recalculated analyses. Base 24 oxygenes				
Si^{4+}	5.827	5.983	5.988	5.495
Ti^{4+}	0.729	0.901	0.922	0.758
Al^{3+}	2.941	2.633	2.562	2.413
Fe^{2+}	3.374	2.602	2.466	2.336
Mn^{3+}	0.024	0.004	0.026	0.0214
Mg^{2+}	2.041	2.695	2.804	2.588
Ca^{2+}	0	0.014	0	0
Na^{+}	0.137	0.095	0.249	0.237
K^{+}	1.912	1.825	1.831	1.582
Ni^{2+}	$2.6x10^{-5}$	0	0	0.005
Zn^{2+}	0	0	0	0
Cr^{2+}	0.007	0.004	0	0
	16.992	16.757	16.849	17.445

the pluton, the absence of metamorphism or deformation, the quite normal biotite's chemical composition, these ages on biotite can be assimilated to crystallization ages. After Odin and Kennedy (1982) this would correspond to the Kimmeridgian (Late Jurassic), or to the Valanginian/Berriasian boundary (Lower Cretaceous) after Harland et al. (1982) and Kent and Gradstein (1985).

Paleomagnetism

Hailwood (1974) by using alternating field demagnetization defined a characteristic direction of the NRM of, D = 315,5, I = -63. As our first purpose was to give an exact age determination for this formation, six huge blocks (between 10 and 10 kg) carefully orientated by sun compass, were sampled within a distance of 2 km. Detailed demagnetizations by A.F. up to 60 mT

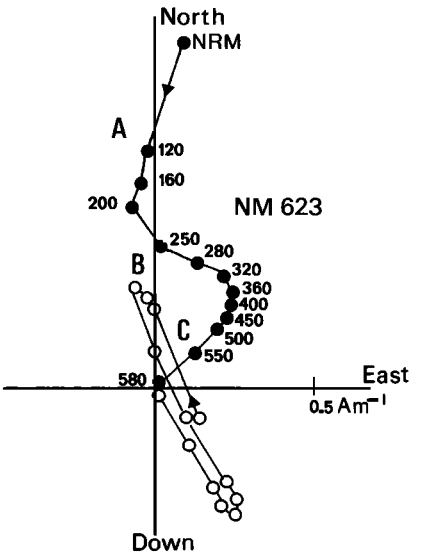


Fig.2. Zijderveld orthogonal vector projections, on the horizontal plane (black dots) and on the vertical E-W (open dots) or N-S (open triangles), Sample NM 623.
Direction A from 0 to 200°C, Direction B, from 200°C to 320°C, which corresponds to the direction isolated by Hailwood (1974), Direction C is from 320°C to 580°C.

and by heating showed that the nature of the magnetizations was far more complex than what was originally found by Hailwood (1974). In two samples, demagnetization has exhibited the direction isolated by Hailwood, but this component is destroyed at 300°C. Another component with higher blocking temperature (600°C) displays a very different direction (Figure 2). In the other blocks different stable directions were found, all characterized by high unblocking temperatures (580-600°C). The erractic results from 31 specimens, preclude to compute any paleopole.

Discussion

Paleomagnetic results on the Msissi norite, either those of Hailwood (1974) or ours differ significantly from conventional Mesozoic directions for Morocco, Westphal et al. (1979), Martin et al. (1978), Hailwood and Mitchell (1971), and does not, therefore, correspond to the K/Ar ages on biotite. Both the freshness of the analysed minerals and the relatively high closure temperature of biotite lead us, however, to regard the

TABLE 2. K/Ar data

SAMPLE	MINERAL	K_2O Weight %	Rad. ^{40}Ar. $\times 10^{-11}$ mol/g	Rad. ^{40}Ar. / Total ^{40}Ar. x 100	Calc. Age ma± σ	Fraction in μm
NM 6	Fr. Biotite	7.66	156.1	93.1	136±4	0.16-0.25
NM 6	Ch. Biotite	4.98	95.1	90.5	128±4	0.1-0.16
NM 6	Fr. Biotite	7.00	145.7	94.4	139±4	0.1-0.16
NM 5	Fr. Biotite	7.58	154.0	96.1	136±4	0.1-0.16

NOTE: $\lambda\varepsilon = 0.581 \times 10^{-11} \times an^{-1}$, $\lambda\beta = 4.962 \times 10^{-10} \times an^{-1}$, $^{40}K/K$ Total $= 1.167 \times 10^{-4}$ mol/mol
Fr. = Fresh, Ch. = Chloritised

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apparent age as representative of crystallization age.

The absence of mechanical or thermal event subsequent to the intrusion strengthens this conclusion. On the other hand, a thermal event able to reset the biotite K/Ar clock would have modified the magnetization, and mostly the lower temperature components. The large scatter of paleomagnetic results indicates that magnetization is far more complex than previously estimated, but does not truly contradict the results of Hailwood's study, simply the components close to 300°C (i.e the direction obtained by Hailwood) can not be older than 140 Ma.

Conclusion

- The term of norite concerning the Msissi dyke is inadequate and we propose that of teschenite (alkaline gabbro).
- The crystallization age of this rock is about 140 M.a. (Upper Jurassic/Early Cretaceous). Therefore the Msissi pole can no more be used as a stable African Devonian paleopole, and the paleoposition of Africa in Devonian times still remains therefore largely unknown.

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